Social Network Analysis Using a Graph Database

1. User Attributes

User	Age	City	Profession	Activity	Influence
Rahul	29	Pune	Data Scientist	92	87
Priya	27	Mumbai	AI Engineer	88	82
Amit	31	Pune	Entrepreneur	76	79
Neha	25	Delhi	Researcher	95	91
Karan	30	Pune	Software Engineer	69	73
Sneha	28	Mumbai	Product Manager	84	86
Arjun	26	Bengaluru	ML Engineer	91	89
Meera	32	Delhi	Consultant	72	77

2. Create the Users

```
CREATE
(r:User {name:'Rahul', age:29, city:'Pune', profession:'Data Scientist', activityLevel:92,
influenceScore:87}),
(p:User {name:'Priya', age:27, city:'Mumbai', profession:'AI Engineer', activityLevel:88,
influenceScore:82}),
(a:User {name:'Amit', age:31, city:'Pune', profession:'Entrepreneur', activityLevel:76,
influenceScore:79}),
(n:User {name:'Neha', age:25, city:'Delhi', profession:'Researcher', activityLevel:95,
influenceScore:91}),
(k:User {name:'Karan', age:30, city:'Pune', profession:'Software Engineer', activityLevel:69,
influenceScore:73}),
(s:User {name:'Sneha', age:28, city:'Mumbai', profession:'Product Manager', activityLevel:84,
influenceScore:86}),
(ar:User {name:'Arjun', age:26, city:'Bengaluru', profession:'ML Engineer', activityLevel:91,
influenceScore:89}),
(m:User {name:'Meera', age:32, city:'Delhi', profession:'Consultant', activityLevel:72,
influenceScore:77});
```

3. Create Friendship Relationships

```
MATCH
(r:User {name:'Rahul'}), (p:User {name:'Priya'}), (a:User {name:'Amit'}), (n:User {name:'Neha'}),
(k:User {name:'Karan'}), (s:User {name:'Sneha'}), (ar:User {name:'Arjun'}), (m:User {name:'Meera'})
CREATE
(r)-[:FRIEND {since:2021, strength:0.91}]->(p),
(r)-[:FRIEND {since:2022, strength:0.75}]->(a),
(r)-[:FRIEND {since:2023, strength:0.64}]->(k),
(p)-[:FRIEND {since:2022, strength:0.83}]->(n),
(p)-[:FRIEND {since:2021, strength:0.89}]->(s),
(n)-[:FRIEND {since:2023, strength:0.71}]->(ar),
(n)-[:FRIEND {since:2022, strength:0.68}]->(m),
(a)-[:FRIEND {since:2024, strength:0.62}]->(k),
(s)-[:FRIEND {since:2023, strength:0.77}]->(ar),
(ar)-[:FRIEND {since:2024, strength:0.66}]->(m);
```

4. Create Groups

```
CREATE
(g1:Group {name:'Artificial Intelligence', category:'Technology', members:420}),
```

```
(g2:Group {name:'Data Science', category:'Technology', members:350}),
(g3:Group {name:'Startup Founders', category:'Business', members:210});
```

5. Connect Users to Groups

```
MATCH
(r:User {name:'Rahul'}), (p:User {name:'Priya'}), (n:User {name:'Neha'}), (a:User {name:'Amit'}),
(s:User {name:'Sneha'}), (ar:User {name:'Arjun'}),
(g1:Group {name:'Artificial Intelligence'}), (g2:Group {name:'Data Science'}),
(g3:Group {name:'Startup Founders'})
CREATE
(r)-[:MEMBER_OF {role:'Member', since:2022}]->(g1),
(r)-[:MEMBER_OF {role:'Moderator', since:2021}]->(g2),
(p)-[:MEMBER_OF {role:'Member', since:2023}]->(g1),
(p)-[:MEMBER_OF {role:'Member', since:2022}]->(g2),
(n)-[:MEMBER_OF {role:'Researcher', since:2021}]->(g1),
(n)-[:MEMBER_OF {role:'Member', since:2023}]->(g2),
(a)-[:MEMBER_OF {role:'Founder', since:2022}]->(g3),
(s)-[:MEMBER_OF {role:'Member', since:2023}]->(g1),
(ar)-[:MEMBER_OF {role:'Member', since:2022}]->(g1),
(ar)-[:MEMBER_OF {role:'Member', since:2023}]->(g2);
```

6. Create Comments

Comments are modeled as independent nodes so each comment can have its own sentiment, toxicity, likes, timestamp and topic.

```
CREATE
(c1:Comment {id:'C001', text:'Graph databases are very useful for connected data.',
sentiment:'Positive', toxicity:0.02, likes:34, timestamp:'2026-08-20'}),
(c2:Comment {id:'C002', text:'I think Graph RAG can improve knowledge retrieval.',
sentiment:'Positive', toxicity:0.01, likes:41, timestamp:'2026-08-21'}),
(c3:Comment {id:'C003', text:'The performance depends heavily on the graph structure.',
sentiment:'Neutral', toxicity:0.03, likes:18, timestamp:'2026-08-21'}),
(c4:Comment {id:'C004', text:'Interesting perspective on community detection.',
sentiment:'Positive', toxicity:0.00, likes:29, timestamp:'2026-08-22'}),
(c5:Comment {id:'C005', text:'I disagree with this approach.', sentiment:'Negative', toxicity:0.12,
likes:9, timestamp:'2026-08-22'});
```

7. Connect Users to Comments

```
MATCH
(r:User {name:'Rahul'}), (p:User {name:'Priya'}), (n:User {name:'Neha'}), (a:User {name:'Amit'}),
(s:User {name:'Sneha'}),
(c1:Comment {id:'C001'}), (c2:Comment {id:'C002'}), (c3:Comment {id:'C003'}), (c4:Comment
{id:'C004'}), (c5:Comment {id:'C005'})
CREATE
(r)-[:COMMENTED {time:'10:30'}]->(c1),
(p)-[:COMMENTED {time:'11:10'}]->(c2),
(n)-[:COMMENTED {time:'12:15'}]->(c3),
(a)-[:COMMENTED {time:'13:20'}]->(c4),
(s)-[:COMMENTED {time:'14:05'}]->(c5);
```

8. Comments as Conversation Nodes

```
MATCH
(p:User {name:'Priya'}), (n:User {name:'Neha'}),
(c1:Comment {id:'C001'}), (c3:Comment {id:'C003'})
CREATE
```

```
(p)-[:REPLIED_TO]->(c1),
(n)-[:REPLIED_TO]->(c1),
(p)-[:REPLIED_TO]->(c3);
```

9. Create Topics

```
CREATE
(t1:Topic {name:'Graph Database'}),
(t2:Topic {name:'Graph RAG'}),
(t3:Topic {name:'Community Detection'}),
(t4:Topic {name:'Social Network Analysis'});

MATCH
(c1:Comment {id:'C001'}), (c2:Comment {id:'C002'}), (c3:Comment {id:'C003'}), (c4:Comment
{id:'C004'}),
(t1:Topic {name:'Graph Database'}), (t2:Topic {name:'Graph RAG'}), (t3:Topic {name:'Community
Detection'})
CREATE
(c1)-[:ABOUT]->(t1),
(c2)-[:ABOUT]->(t2),
(c3)-[:ABOUT]->(t1),
(c4)-[:ABOUT]->(t3);
```

13. Important Cypher Patterns

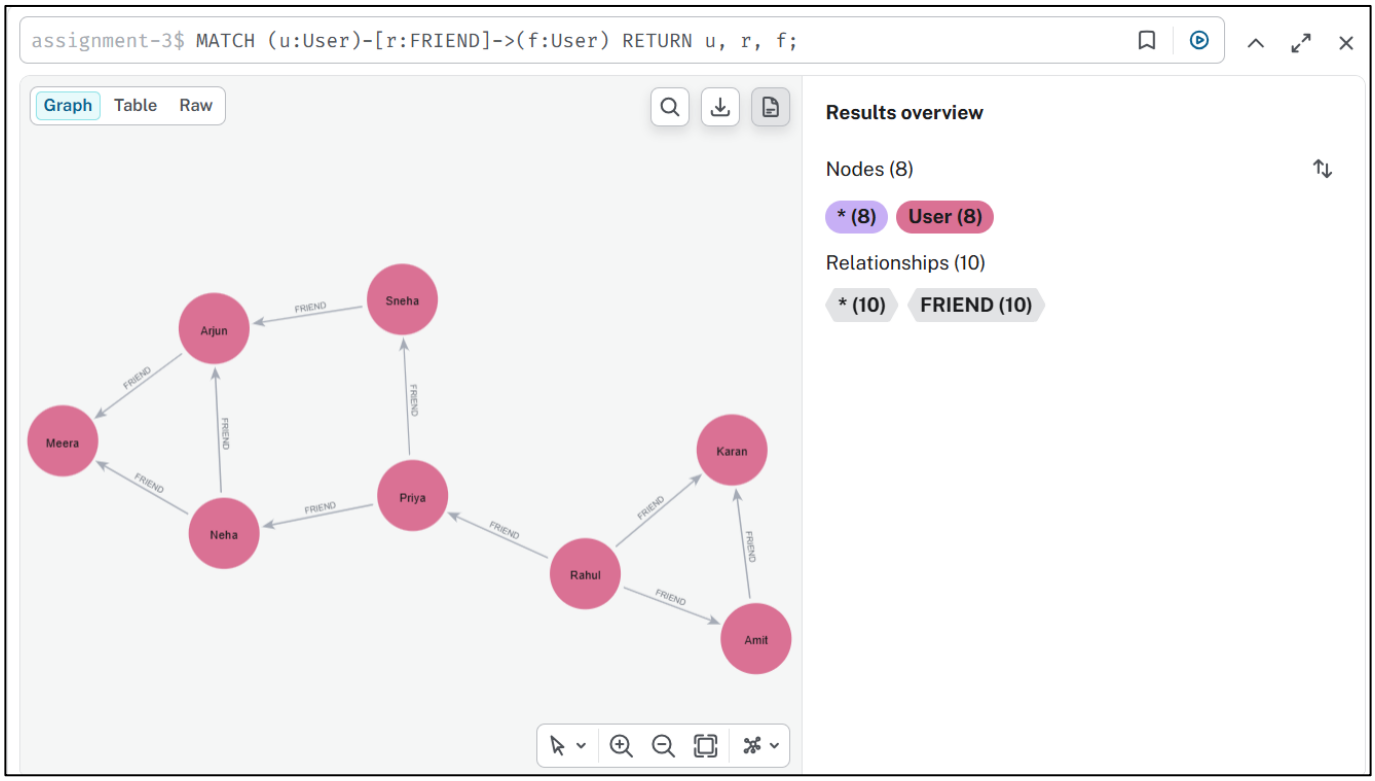
Operation	Pattern
Create node	CREATE (u:User {...})
Find node	MATCH (u:User {...})
Create relationship	CREATE (a)-[:FRIEND]->(b)
Find relationship	MATCH (a)-[r:FRIEND]->(b)
Any relationship	MATCH (a)-[r]-(b)
Variable-length path	MATCH path=(a)-[:FRIEND*1..3]-(b)
Return graph	RETURN a, r, b
Return path	RETURN path
Filter	WHERE condition
Count	COUNT(x)
Update	SET u.property = value
Delete relationship	DELETE r
Delete node + relationships	DETACH DELETE u

12. Graph Visualization

Purpose	Cypher	Typical result
Tabular	MATCH (u:User)-[:FRIEND]->(f) RETURN u.name, f.name	Names/rows
Graph	MATCH (u:User)-[r:FRIEND]->(f) RETURN u,r,f	Nodes + edges
Path	MATCH path=(u)-[:FRIEND*1..3]-(v) RETURN path	Connected path

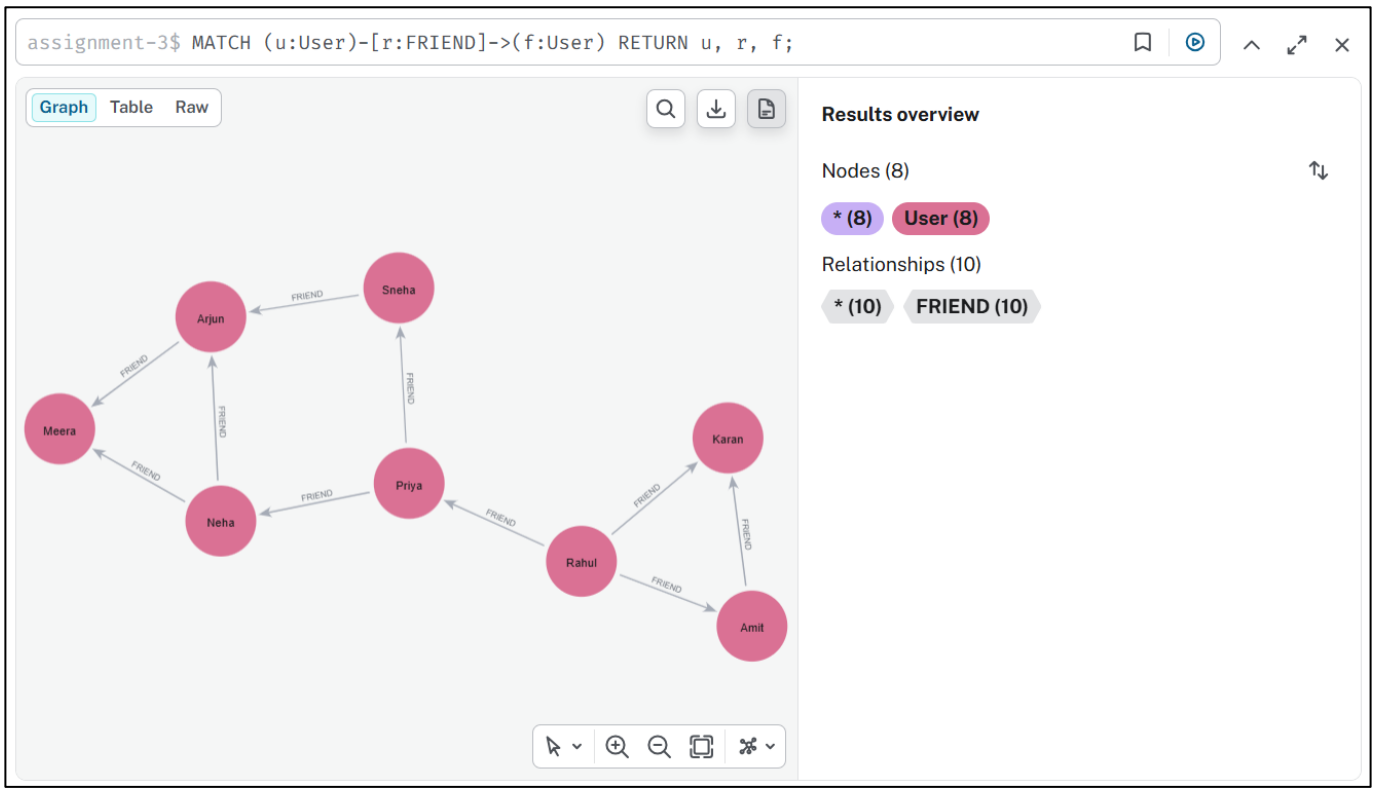
Exercise 1 — Display the Complete User Network

Returns a visual network containing users and user-to-user relationships.



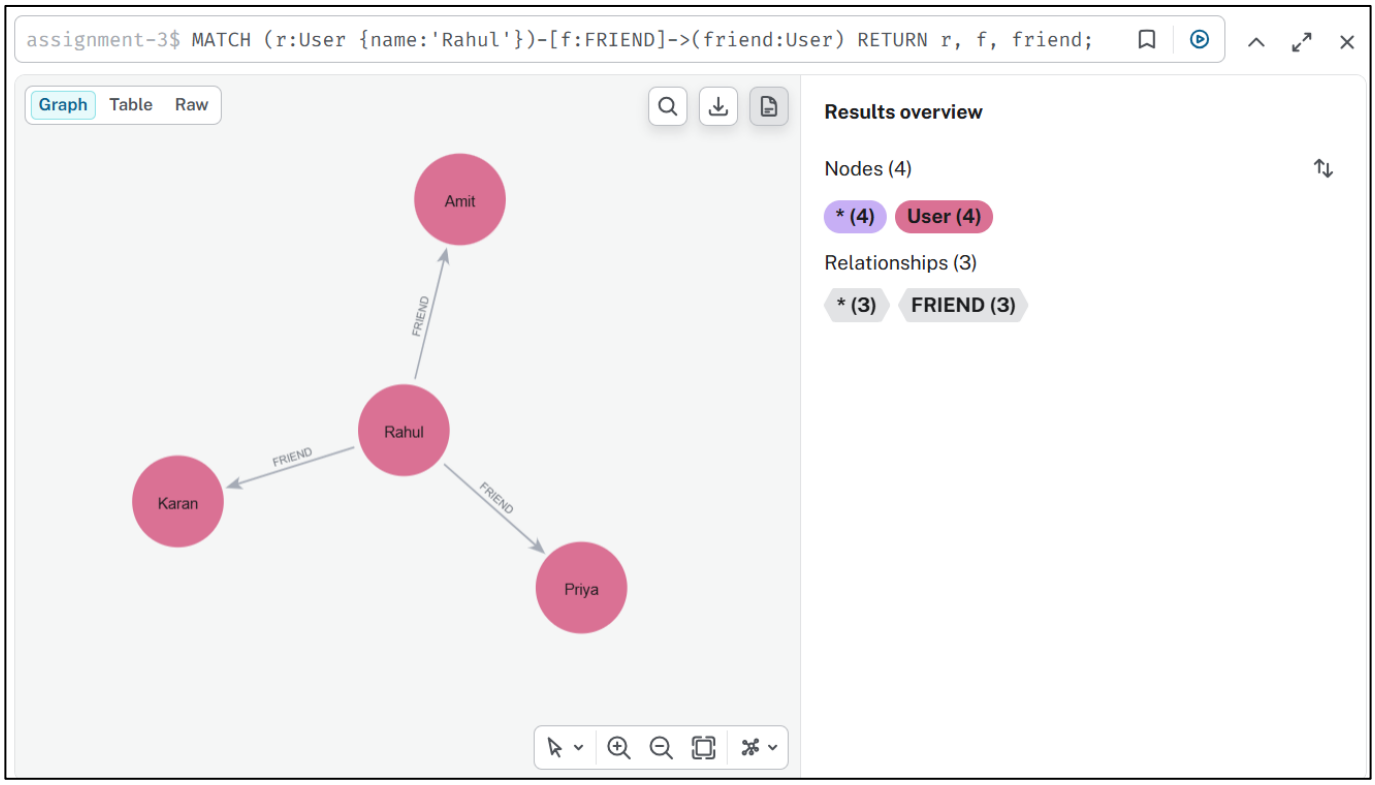
Exercise 2 — Display Only Friendships

Shows the friendship graph.



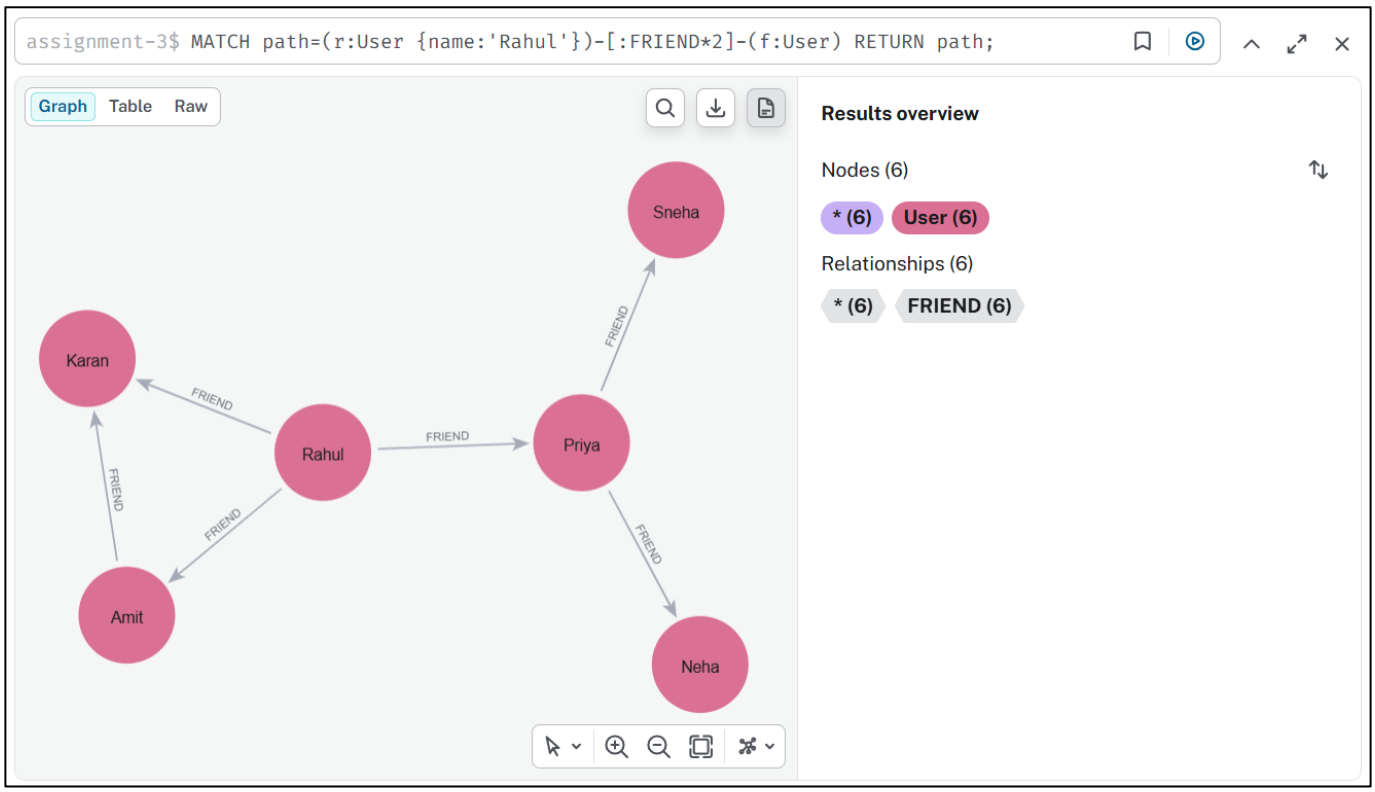
Exercise 3 — Rahul's Immediate Network

Builds Rahul's one-hop ego network.



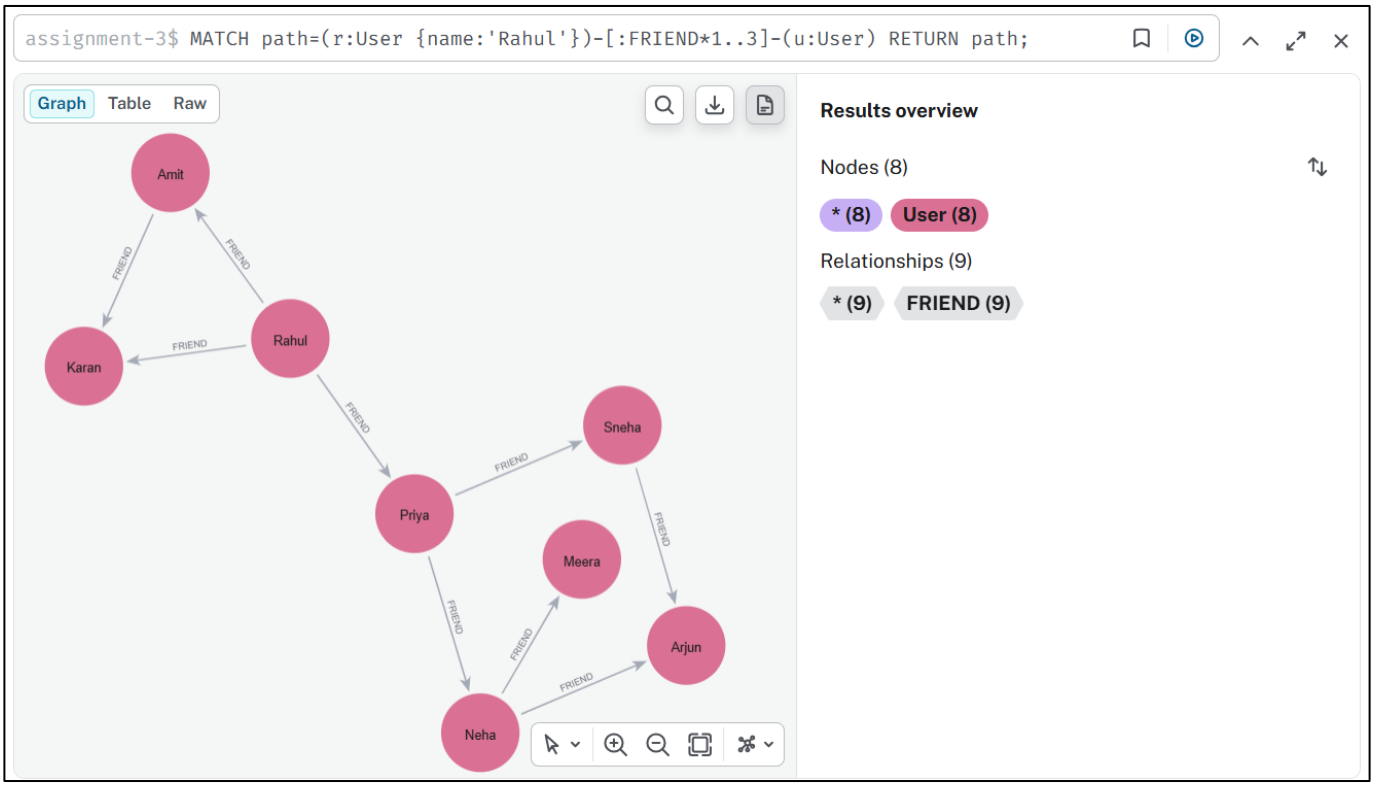
Exercise 4 — Rahul's Friends-of-Friends

Visualizes exactly two friendship hops.



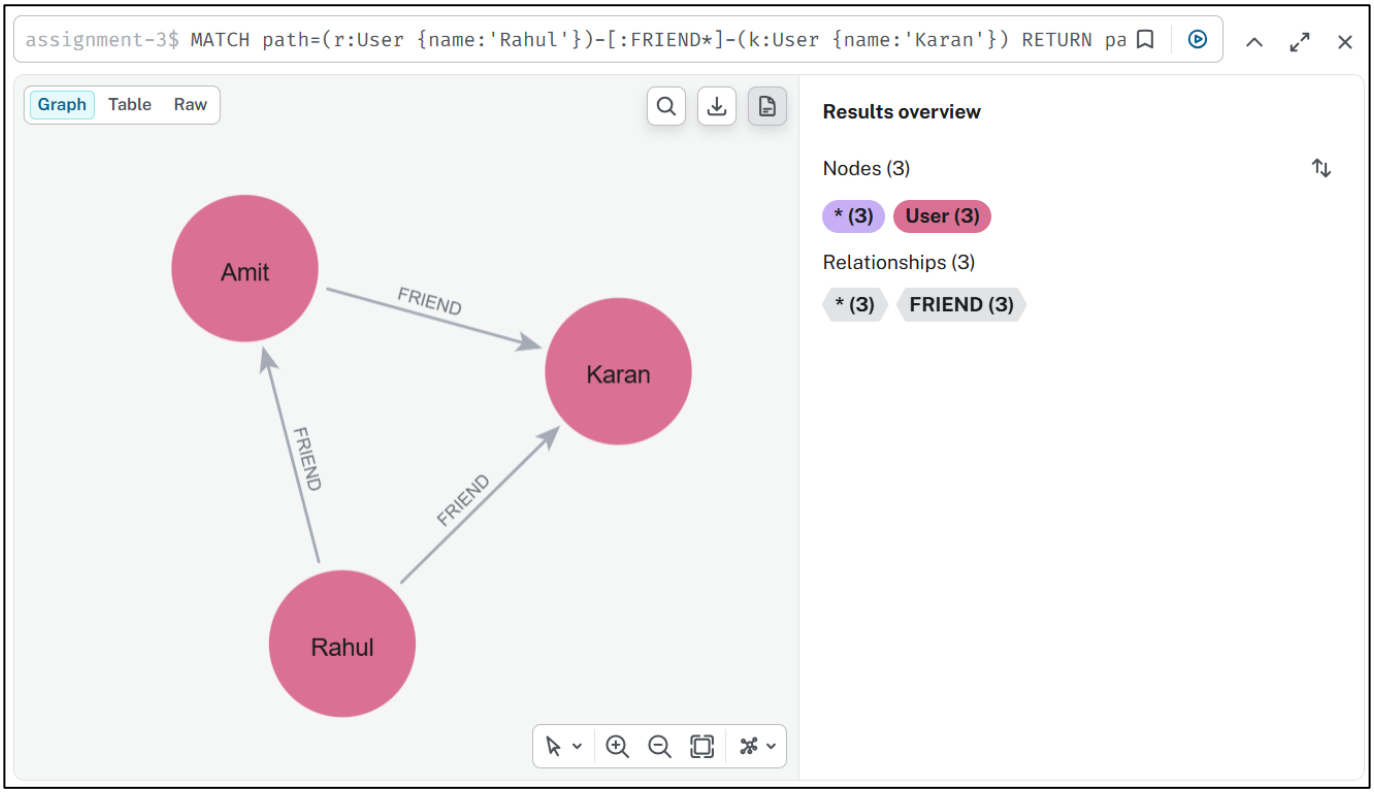
Exercise 5 — One to Three Hops

Visualizes Rahul's wider social neighborhood.



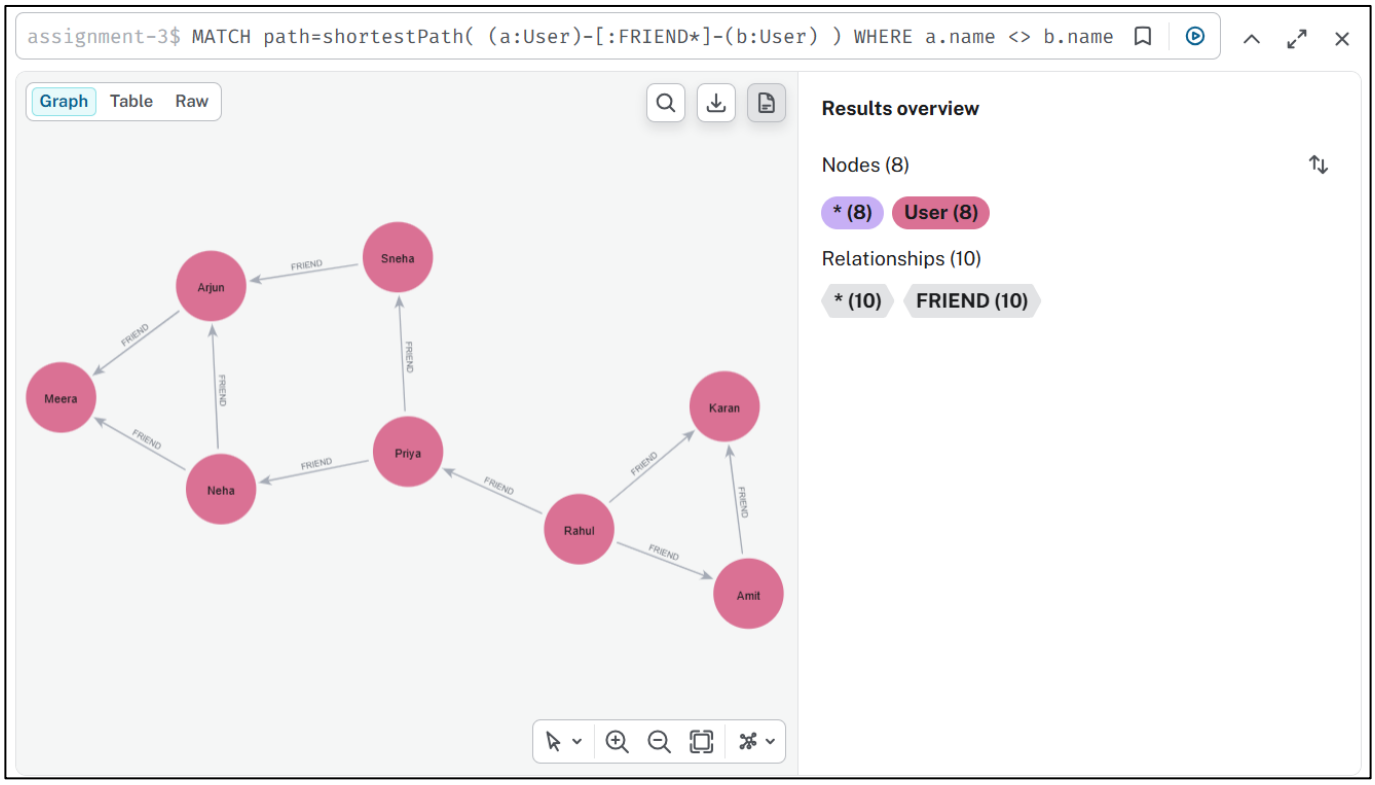
Exercise 6 — Connection Between Rahul and Karan

Returns friendship paths connecting the two users.



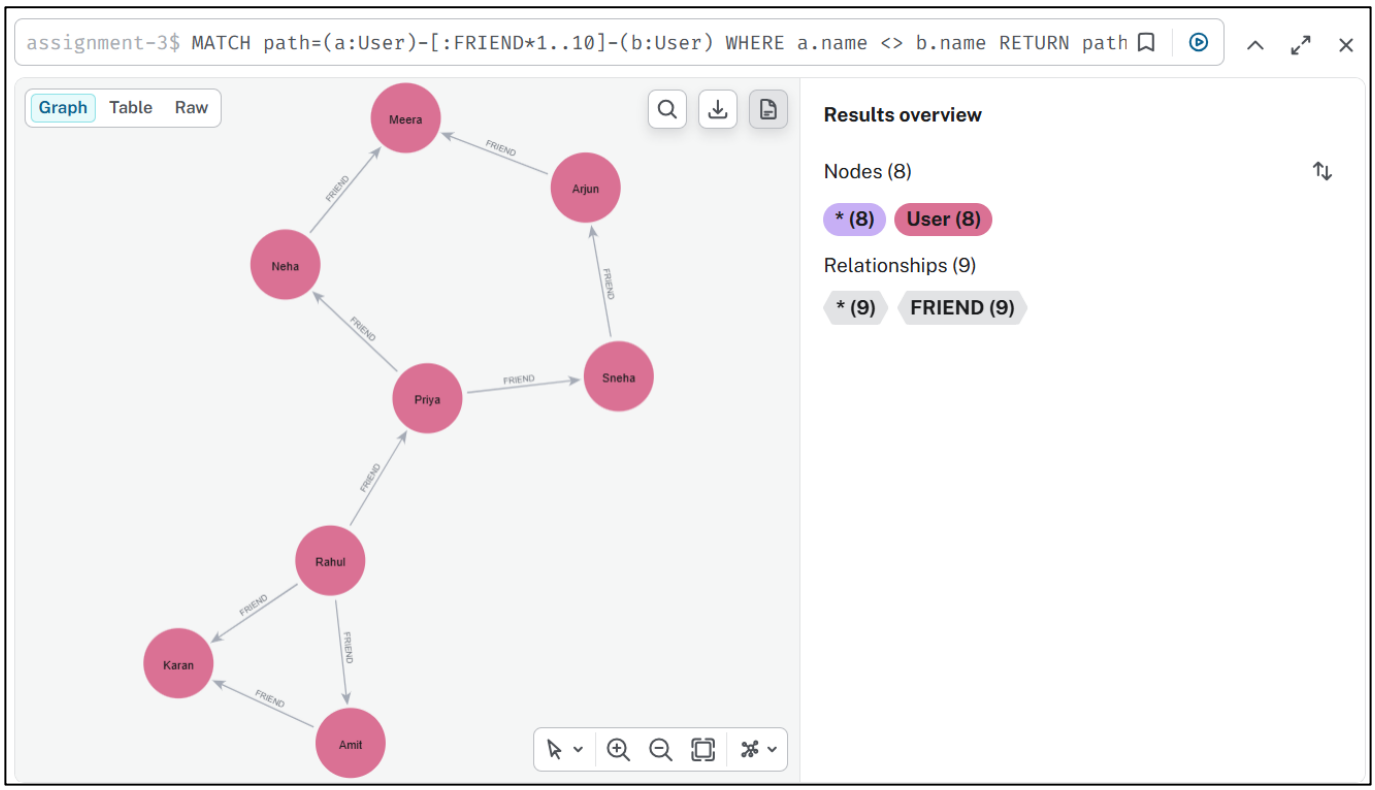
Exercise 7 — Shortest Social Connection (Any user)

Finds the shortest friendship route.



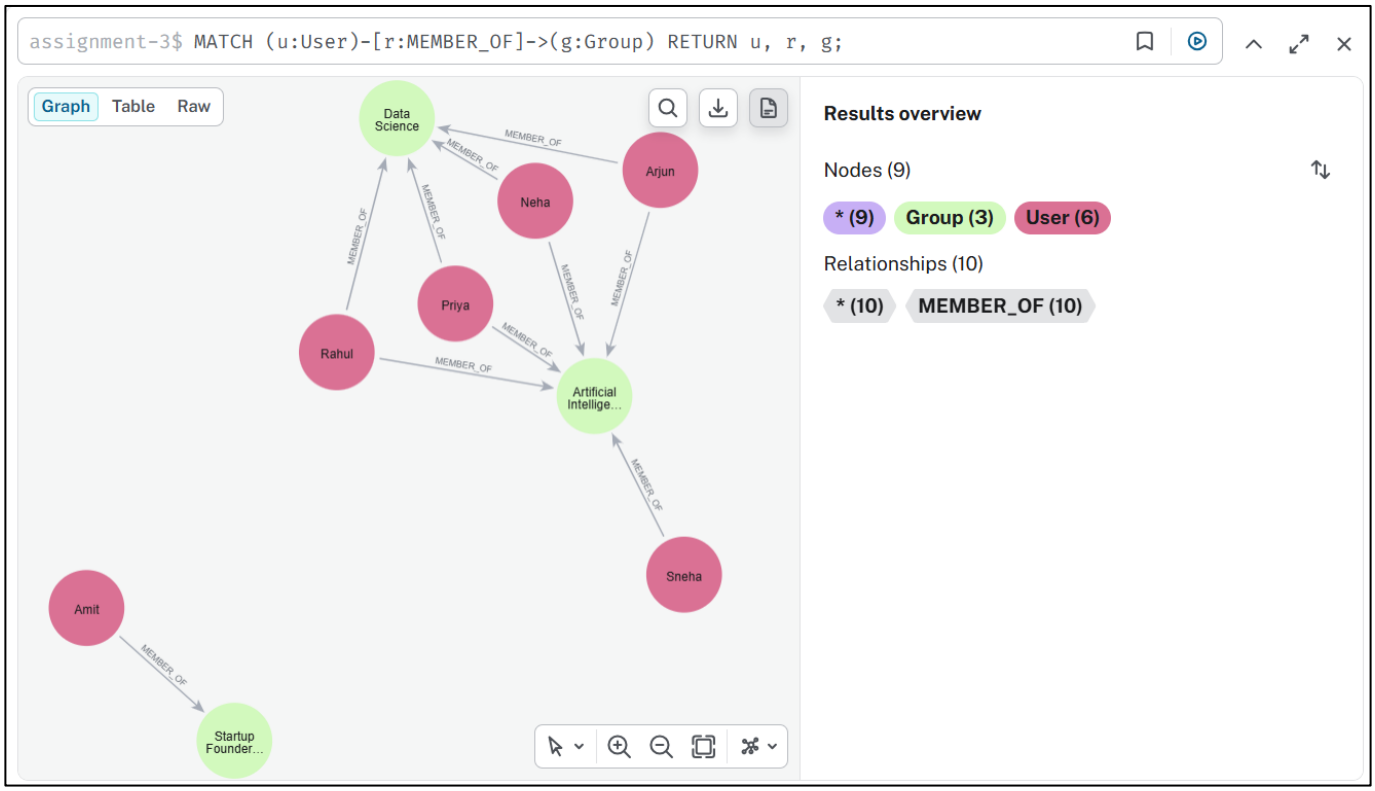
Exercise 8 — Longest Social Connection

Finds the shortest friendship route.



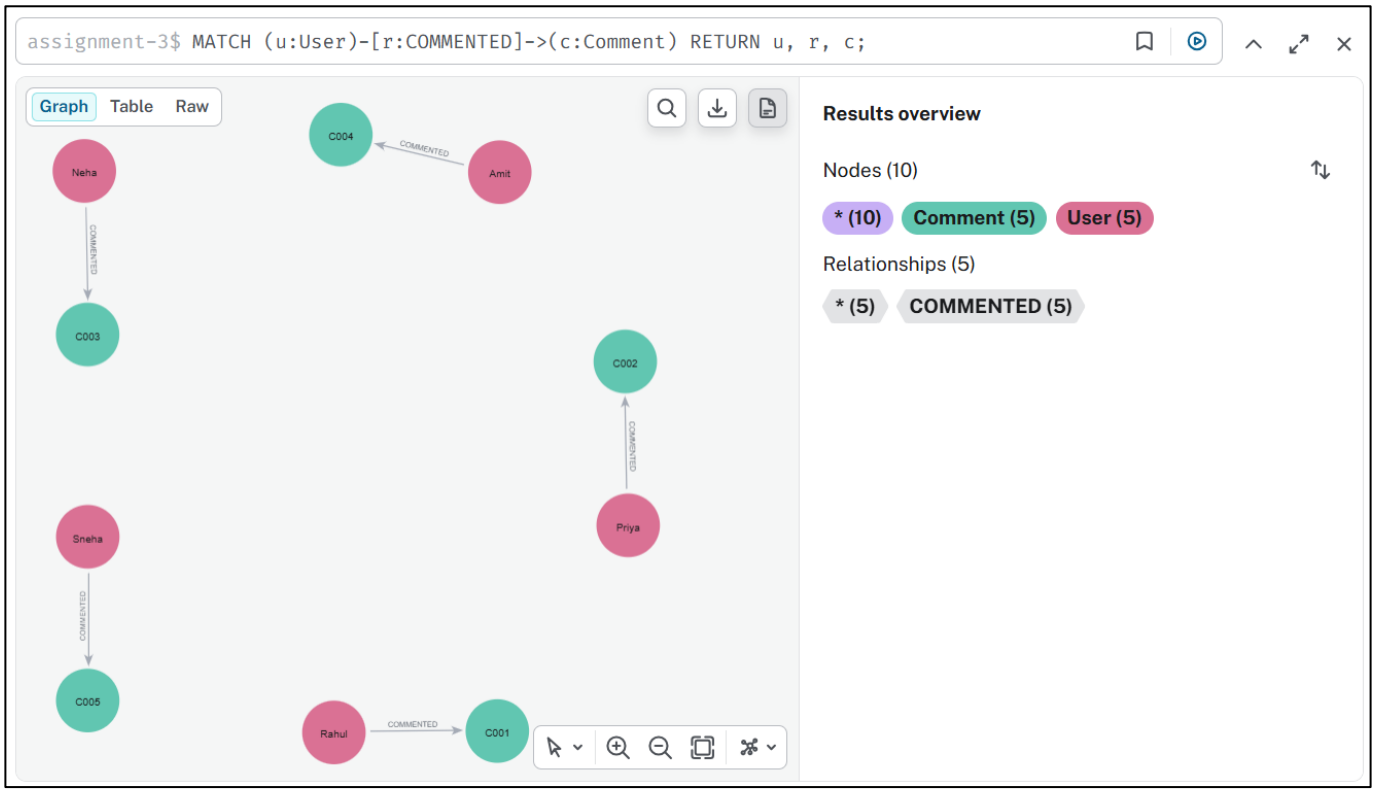
Exercise 9 — Users and Groups

Visualizes community/group membership.



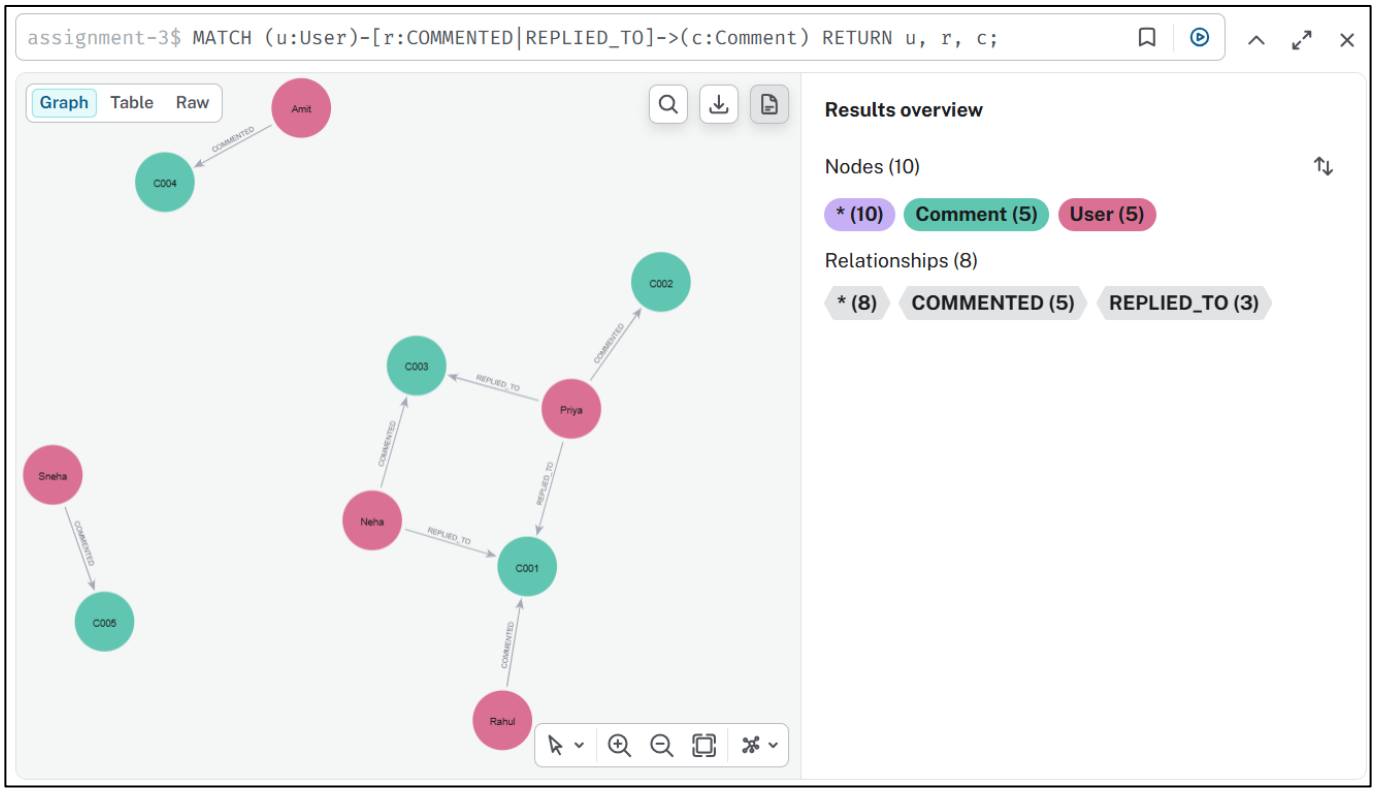
Exercise 10 — Users and Their Comments

Visualizes who made which comment.



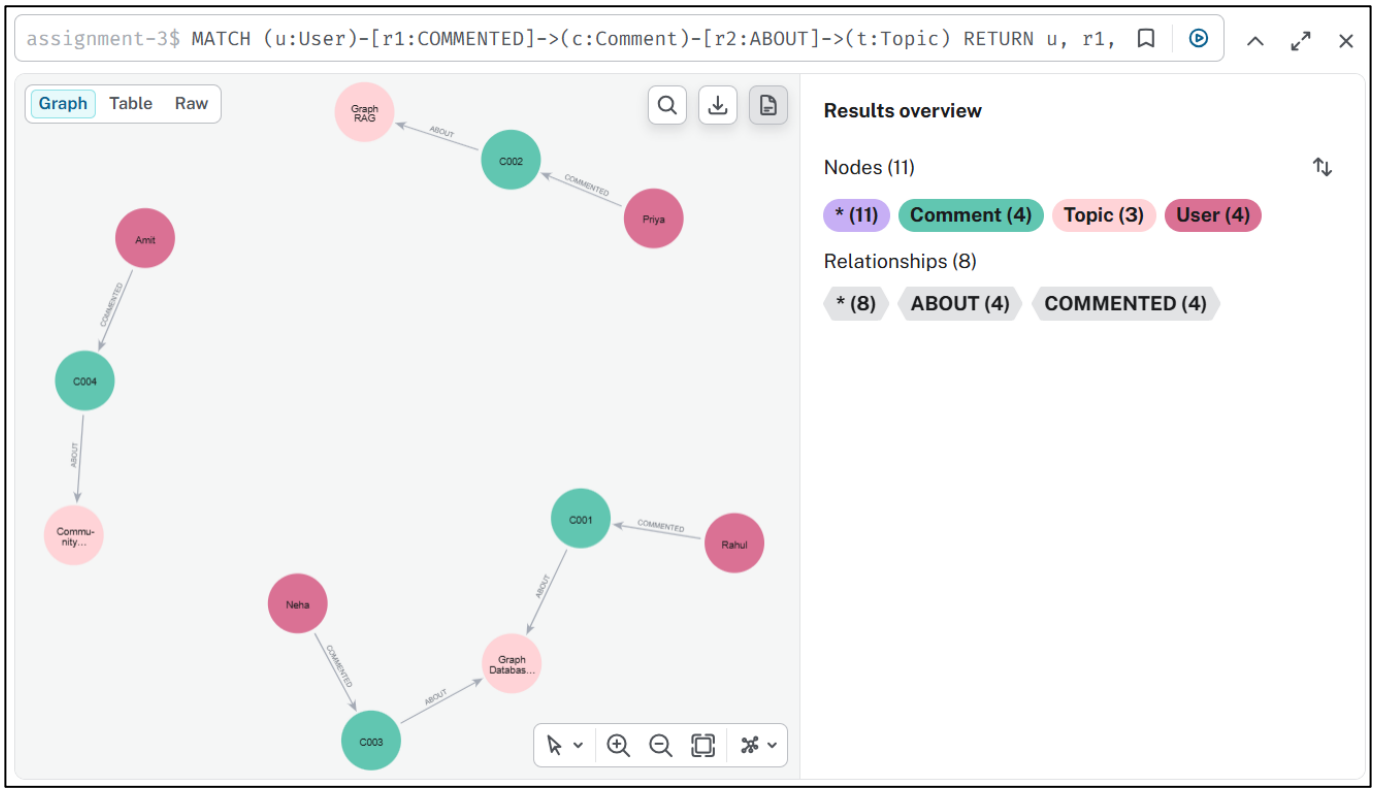
Exercise 11 — Comment Conversation Network

Shows users participating in a comment conversation.



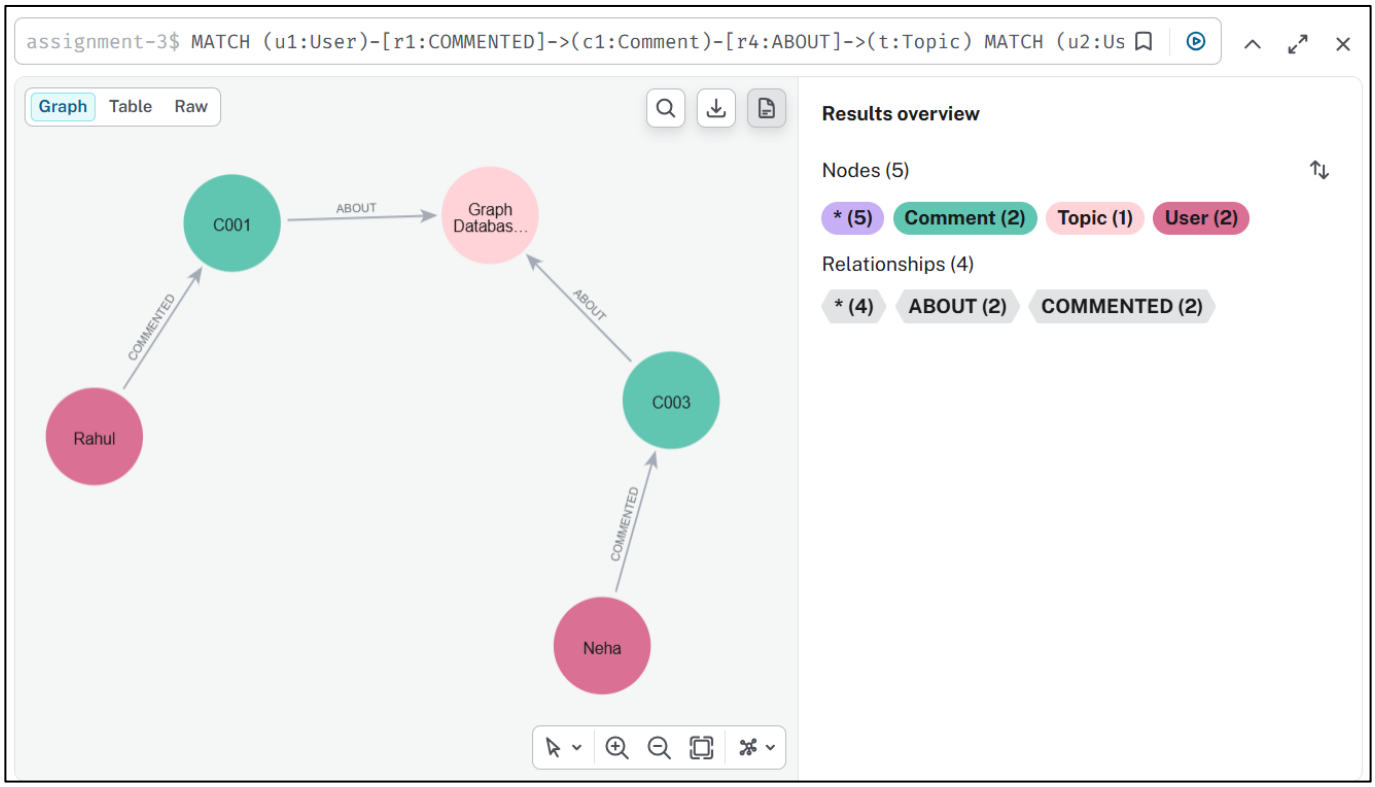
Exercise 12 — Users, Comments and Topics

Connects people to topics through comments.



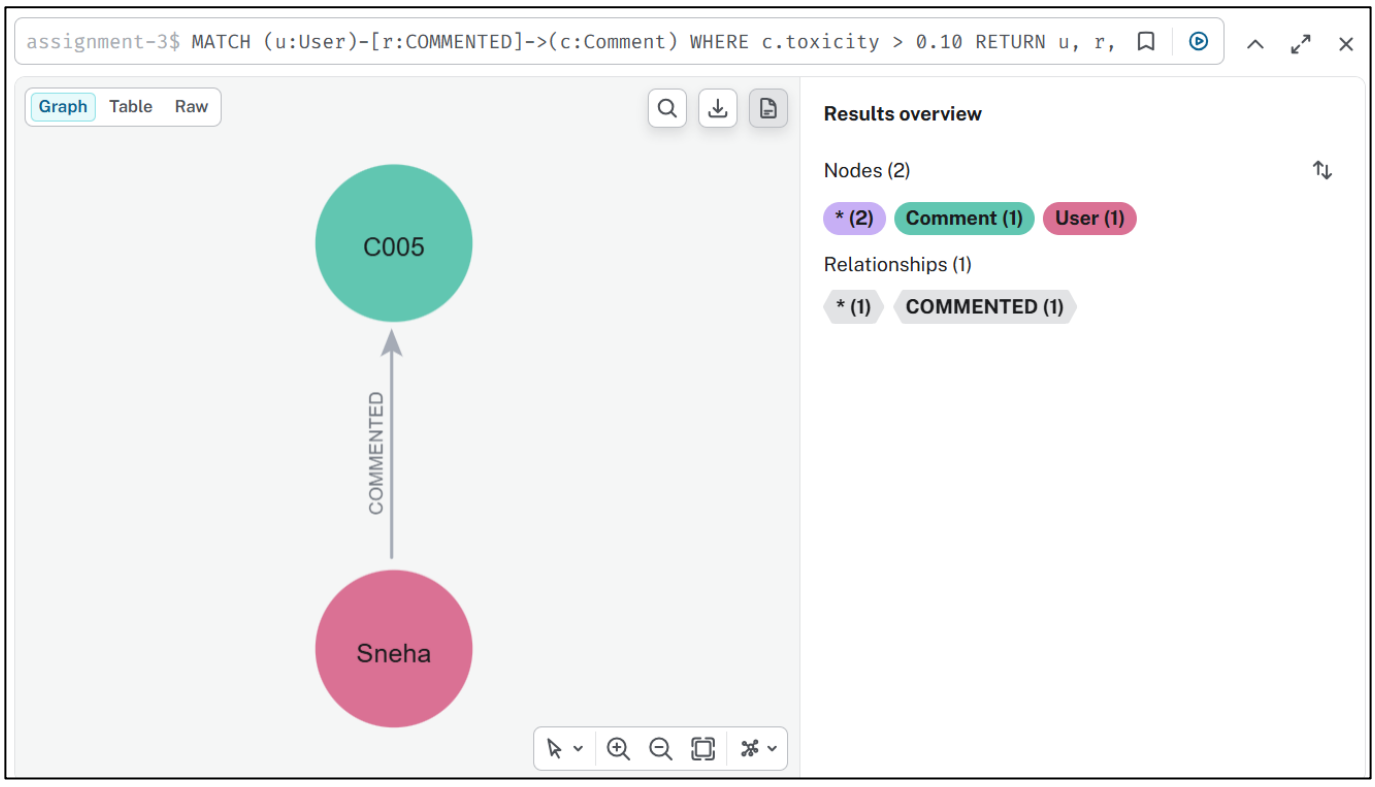
Exercise 13 — Shared Topic Network

Visualizes users discussing the same topic.



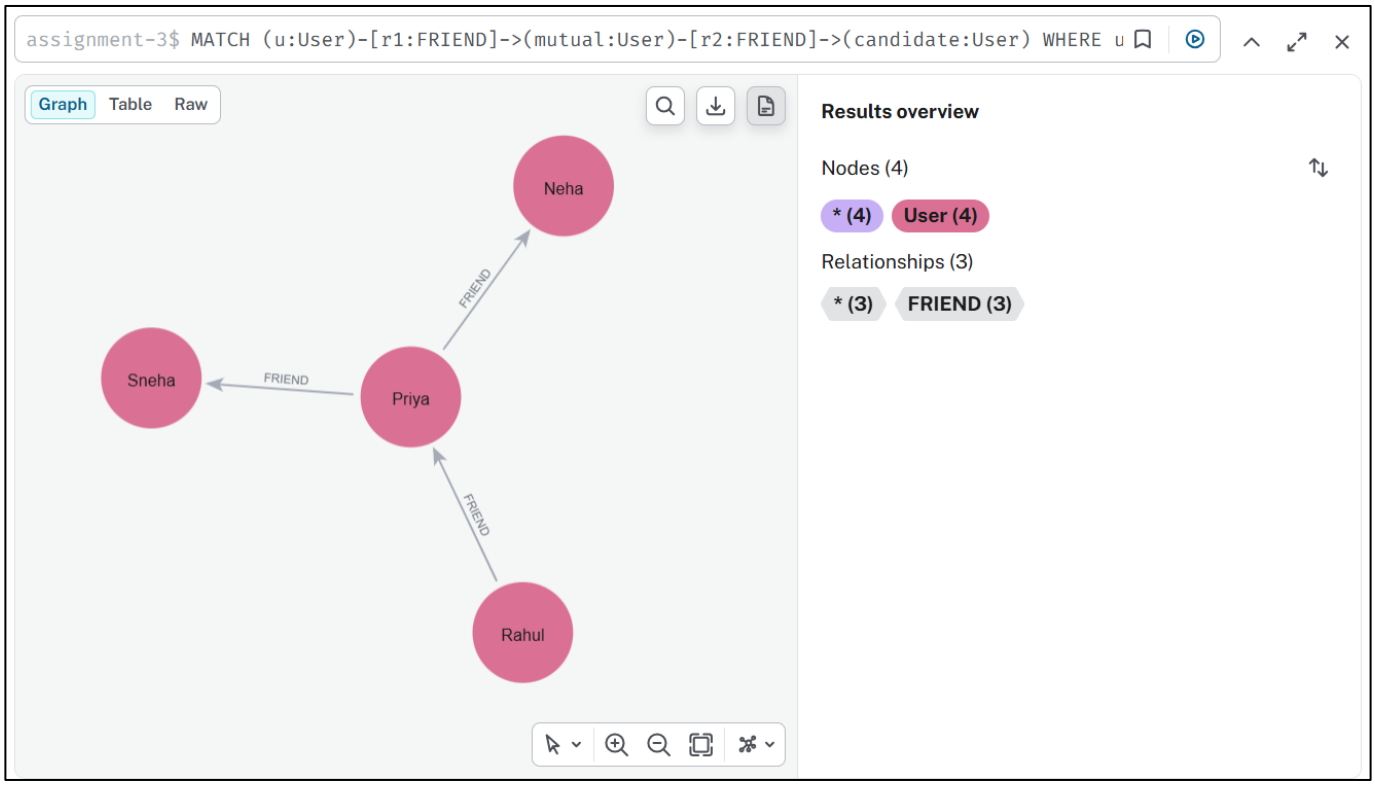
Exercise 14 — High-Toxicity Interaction

Visualizes conversations around high-toxicity comments.



Exercise 15 — Friend Recommendation Graph

Returns the graph structure behind a friend recommendation.



Exercise 16 — Complete Social Interaction Graph

Visualizes all currently modeled relationships.

